

ADTA 5750.501: Applied Natural Language Processing

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1. Overview

The midterm covers all the topics discussed in the first half of the course. Materials in any format, including in-class discussion, should be considered and used for the midterm. Additionally, the student can use any other source of information that he/she can gather, provided it is relevant and supports the student's answers.

The student must create an MS Word document named “**ADTA5750_midterm.docx**” that contains all their midterm work, except for the Python coding.

IMPORTANT NOTES:

- > *The student should present their work for each section using both text and images.*
- > *The sources can be from class lectures, assignments, etc., or other sources*
- > *One picture is worth 1000 words. Adding one or more images along with the text would be much more convincing. However, an image with text explaining what it is and what it is for is considered complete.*
- > *Images can include the screenshots that the student has taken while working on the classwork.*

IMPORTANT NOTES:

- > *When discussing a topic or answering a question, the student must provide adequate explanation and supporting details to their presentation.*

IMPORTANT NOTES:

- > *If MS Docx is the document format required for submission, the student must submit the contents as MS Docx files, **not** submit PDF documents.*

IMPORTANT NOTES:

- > *If Jupyter Notebook documents are required in a Submission Requirement, the student must **submit the Jupyter Notebook documents**, i.e., ***.ipynb documents**, **not** python programs, i.e., **not** *.py programs.*

IMPORTANT NOTES:

- > *For the Python code, the student must write the **code of each step** in **one cell** of the Jupyter Notebook document, as shown in the lectures. Then the student must **run the code of each cell to show the results of each step** in each submitted Jupyter Notebook document.*

2. PART I: Short Questions and Answers (25 Points)

IMPORTANT NOTES:

--> The student can answer the following questions in PART I using the lectures or the contents of the GCP Dialogflow CX documentation.

--> The student should **not** search the Internet (except the GCP Dialogflow CX documentation) for the answers to the questions. The information provided by the Internet (except the GCP Dialogflow CX documentation) may not be correct or suitable.

IMPORTANT NOTES: HOW TO ANSWER EACH QUESTION

--> Each question is followed by the text “ANSWER” that marks where the answer to the question should be entered.

--> The student answers each question by typing the answer directly after the mark “ANSWER.”

It is assumed that an **AI dialogue system** has been **designed** and **deployed** using the GCP **Dialogflow CX** following the concepts and steps discussed in the GCP: Dialogflow CX documentation.

Answer the following questions with short answers:

1. A dialogue/conversation can be represented and visualized with a state machine: TRUE or FALSE?

ANSWER:

2. If so (from QUESTION 1), what concept(s) of Dialogflow CX can be matched with a state of the above state machine?

ANSWER:

3. Provide an example of the pair: ENTITY TYPE – LIST OF ENTITY ENTRIES

ANSWER:

4. Provide an example of mapping entities

ANSWER:

5. Describe different types of state handlers

ANSWER:

6. What is a route?

ANSWER:

7. What is an event handler?

ANSWER:

8. If a route is applied to a flow and the current page of the flow is the Default Start Page, is the route in scope?

ANSWER:

9. What routes can be evaluated?

ANSWER:

10. When handlers are evaluated, which type of handlers is evaluated first?

ANSWER:

11. How many pages can be “active” at any moment along a flow of conversation?

ANSWER:

12. What are entity types used for?

ANSWER:

13. How does Dialogflow find a match in the intent matching process?

ANSWER:

14. What does Dialogflow try to find in the intent matching: A 100% match or a best match?

ANSWER:

15. What are training phrases of an intent?

ANSWER:

16. Do we need to define all possible training phrases of an intent?

ANSWER:

17. Following the best practices , how many training phrases should be created for each intent, following the best practices?

ANSWER:

18. What are the types of events? How are events identified?

ANSWER:

19. How many steps must be taken to process a state handler? List all of them in the order.

ANSWER:

20. What scopes can a route be applied to?

ANSWER:

21. What scopes can event handlers be applied to?

ANSWER:

22. If a route is applied to a page, how can it be determined to be in scope or not?

ANSWER:

23. When are built-in events invoked?

ANSWER:

24. What is a custom event?

ANSWER:

25. How many kinds of data can a state handler have? List all of them.

ANSWER:

26. How many types of state handlers are there?

ANSWER:

27. When can state handlers be evaluated?

ANSWER:

28. When can state handlers be called?’

ANSWER:

29. Each flow must have at least a page: TRUE or FALSE

ANSWER:

30. How many phases does the life cycle of a page have? List all of them in the order.

ANSWER:

31. What happens to the intent, i.e., intent requirement when an intent route is called?

ANSWER:

32. What happens to the condition, i.e., condition requirement when a condition route is called?

ANSWER:

33. What happens to the event, i.e., event requirement when an event handler is called?

ANSWER:

34. How is the scope of a state handler (a route or an event handler) determined?

ANSWER:

35. Provide an example of list entities

ANSWER:

36. Provide an example of composite entities

ANSWER:

37. Provide an example of regexp entity

ANSWER:

38. At least two intents are automatically created when the agent is created. What are they

ANSWER:

39. What major entity options does GCP Dialogflow CX support?

ANSWER:

40. Can multiple messages be added to a response queue?

ANSWER:

41. Provide an example of a negative training phrase that can be used to train an agent on some intent.

ANSWER:

42. What are two built-in event handlers that are automatically created for any flow?

ANSWER:

43. Why do we need pages?

ANSWER:

44. When is a route called?

ANSWER:

45. What happens if an intent matching process finds that no intents meet the classification threshold setting?

ANSWER:

46. Can the NO-MATCH event handler of a flow be deleted?

ANSWER:

47. What is intent propagation? Provide one example to illustrate what happens when intent propagation occurs.

ANSWER:

48. Can multiple pages have a common set of routes?

ANSWER:

49. How to change the current page to a specific page in another flow?

ANSWER:

50. A form parameter is also called a re-prompt handler: TRUE or FALSE?

ANSWER:

3. PART II: Develop Conversational AI System for G-Records (45 Points)

IMPORTANT NOTES:

--> The student must **finish HW 3 before** continuing working on PART II of the midterm.

IMPORTANT NOTES:

--> For the following tasks, the student should take a screenshot of each step of a task and use the screenshots for illustration in the report.

--> For each TO-DO task, the student **only needs to take screenshots of the steps** of a process **once**. **For example**, the student needs to add three dialog flows. The student only needs to capture the screenshots of the steps of the process to add the first dialog flow.

TO-DO #1

- Access and read the tutorials on GCP: Dialogflow CX to build a virtual retail agent
 - At this link
 - <https://codelabs.developers.google.com/codelabs/dialogflow-cx-retail-agent#0>

TO-DO #2

- Following the steps discussed in the tutorial, complete the steps to set up pages and state handlers.

TO-DO #3

- Next, complete the steps to add conditional responses.

TO-DO #4

- Next, complete the steps to wrap up the virtual retail agent.

TO-DO #5

- Finally, complete the steps to test the virtual retail agent.

TO-DO #6

- While working on each TO-DO of PART II, capture the screenshots of each step.
 - **NOTES:** Completing each TO_DO requires multiple steps.
- Write a report for each above section (**TO-DO 2 – 5**) with details about what has been done at each step, illustrated with the screenshots.

SUBMISSION REQUIREMENTS PART II:

--> Submit the report, including the details of each step and the screenshots.

4. PART III: In Class (Monday 10/07/2024) (30 Points)

IMPORTANT NOTES:

--> The student completes **PART III** of the MIDTERM **in class** on **Wednesday 10/07/2024**.

5. HOWTO Submit (PART I and PART II)

Due date & time: 11:00 PM – Wednesday 10/16/2024

The student must submit the midterm (PART I and II) in a Microsoft Word document as an attachment to a UNT email that is sent to the instructor (Thuan.Nguyen@unt.edu).

The subject of the email must be: “**ADTA 5750: Midterm Assessment – Submission.**”